Multi-Robot Path Planning with Efficiently Adaptive State Lattices

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Problem Statement

Autonomous robots operating in multi-robot teams require path planning algorithms that can plan efficient and safe routes through complex, dynamic environments. For mobile robots with nonholonomic constraints and non-circular footprints, motion planning algorithms such as Efficiently Adaptive State Lattices (EASL) [Hedegaard 2021] can be used to generate trajectories that avoid collisions with static and dynamic obstacles. EASL routes predict when the robot will reach a state at a particular time, which is beneficial for multi-robot motion planning where routes often overlap in space but not necessarily at the same time. In this research, we investigate algorithms based on EASL for robot motion planning than enable robots to quickly plan routes through narrow passageways without approximating the footprint of the vehicle as a circle.

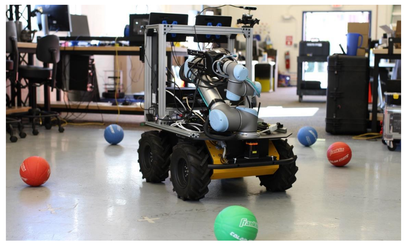


Figure 1: Clearpath Robotics Husky A200 Mobile Robot

Background

EASL was introduced in [Hedegaard 2021], which use a discrete approximation of the continuous adaptive state space explored in [Howard 2009, Napoli 2018]. Connectivity is generated in EASL using a model-predictive trajectory generation algorithm introduced in [Howard 2007], which uses a parametric approximation of continuous inputs to iterative refine a initial guess of parametric inputs until boundary state constraints are satisfied. Environments were represented as binary costmaps in [Hedegaard 2021] and later extended to non-flat terrain in [Damm 2023].



Figure 2: Model-predictive trajectory generation from [Howard 2009]

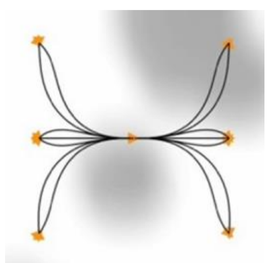


Figure 3: State lattice control set from [Howard 2009]

Technical Approach

A trajectory generator based on [Howard 2009] was implemented that satisfies a two-point boundary value problem for edges in a recombinant state lattice.

$$\Delta\mathbf{x} = \begin{bmatrix} x_d - x_f(\mathbf{p}) & y_d - y_f(\mathbf{p}) \end{bmatrix}^T$$

Difference between the desired position x_d, y_d and initial position $x_f(\mathbf{p}), y_f(\mathbf{p})$

$$J(\mathbf{p}) = \begin{bmatrix} \dfrac{x_f(\mathbf{p}, \phi_1 + \eta) - x_f(\mathbf{p})}{\eta} & \dfrac{x_f(\mathbf{p}, t_f + \eta) - x_f(\mathbf{p})}{\eta} \\ \dfrac{y_f(\mathbf{p}, \phi_1 + \eta) - y_f(\mathbf{p})}{\eta} & \dfrac{y_f(\mathbf{p}, t_f + \eta) - y_f(\mathbf{p})}{\eta} \end{bmatrix}$$

Jacobian Approximation($J(\mathbf{p})$): This matrix computes the sensitivity of the final x and y positions to a small perturbation (η) which is applied to the parameter space p = [ϕ_1 tF]^T.

$$\Delta\mathbf{p} = J(\mathbf{p})^{-1}\Delta\mathbf{x}$$

Parameter Correction (Δp): Computes the required correction to the trajectory parameters through multiplying the Jacobian and the position error.

EASL uses Anytime Repairing A* for heuristic search to find an optimal path through a binary representation of an obstacle field subject to kinematic constraints and avoid obstacle that overlap with the vehicle's footprint. Rather than assume that the rectangular shaped vehicle (shown in Figure 1) is circular or continuously check the vehicle footprint against the environment map as was done in [Hedegaard 2021], we construct a three-dimensional representation of the two-dimensional obstacle map by sampling various discrete headings in SE(2) and precomputing obstacle collisions in this three-dimensional space. This is achieved by creating a "stamp" of nearby states at various headings that collide with an obstacle and iterating over the two-dimensional map to populate the three-dimensional obstacle map. A state lattice planner based on EASL was implemented that used this representation for collision checking in this project.

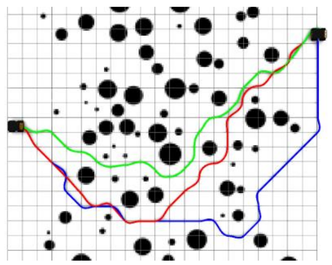


Figure 4: SL, DASL, and EASL solutions from [Hedegaard 2021] showing paths that avoid obstacles.

Preliminary Results

The current pipeline was validated in a simulated 2D occupancy grid populated with a custom map with static circular obstacles and a start and target goal locations. Figure 5 and 6 demonstrates the kinematic constraints and the 3D (X, Y, θ) collision map.

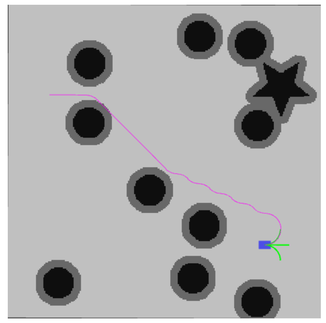


Figure 5: EASL Solution Showing Path to goal with robot's initial orientation facing away from the target goal

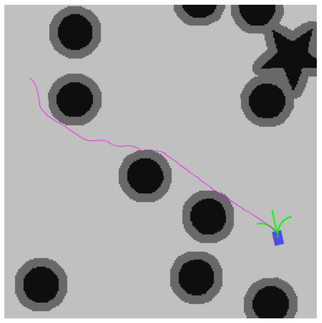


Figure 6: EASL Solution Showing Path to goal around with the robot's initial orientation facing toward the target goal

Future Work

The next step of this research project involves applying the fast three-dimensional collision map for multi-robot motion planning for generating trajectories through obstacle fields composed of static and dynamic obstacles, where some dynamic obstacles are other robots that we plan routes for. We can also explore jointly searching in the higher dimensional space of multi-robot team actions, through the computational complexity of such search increases exponentially. Further research will test the efficacy of these algorithms in a multi-robot scenario where robots are tasked with actions that involve operating in a shared workspace.

Acknowledgements

I would like to express my gratitude to my Principal Investigator, Professor Thomas Howard, for his advice and guidance. I also thank Ph.D. student Tabib Wasit Rahman for his assistance with software debugging. Finally, thank you to the generous donors who made this research grant possible.

Thomas M. Howard and Alonzo Kelly. Optimal rough terrain trajectory generation for wheeled mobile robots. International Journal of Robotics Research, 26(2):141–166, February 2007

Thomas M. Howard. Adaptive Model-Predictive Motion Planning for Navigation in Complex Environments. PhD thesis, Carnegie Mellon University, August 2009

Michael Napoli, Harel Biggie, and Thomas M. Howard. Learning models for predictive adaptation in state lattices. In Field and Service Robotics: Results of the 11th International Conference. Springer Proceedings in Advanced Robotics, volume 5, pages 285–300. Springer, Cham, 2018

Benned Hedegaard, Ethan Fahnestock, Jacob Arkin, Ashwin Menon, and Thomas M. Howard. Discrete optimization of adaptive state lattices for iterative motion planning on unmanned ground vehicles. In 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 5764–5771, 2021

Eric R. Damm, Jason M. Gregory, Eli S. Lancaster, Felix A. Sanchez, Daniel M. Sahu, and Thomas M. Howard. Terrain-aware kinodynamic planning with efficiently adaptive state lattices for mobile robot navigation in off-road environments. In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 9918–9925, 2023